1. Understand and utilise constructs based on definitions... (Point 5)

- **Analysis:** This is the core of solving the problem. Students must understand geometric definitions (the relationship between diameter and radius) and the rules they follow in physical space. To find the height of the center M, students need to use the basic rule that "the radius is half the diameter" (\(140 \div 2 = 70\)) and apply it to the given spatial structure.

2. Manipulate numbers, graphical data... and geometric representations (Point 7)

- **Analysis:** The problem provides a geometric diagram (circle, tangent line, vertical height). Students need to manipulate this graphical data: either subtract the radius (70m) from the highest point's height (150m), or add the radius (70m) to the lowest point's height (\(150 - 140 = 10\text{m}\)). This manipulation between the geometric representation (line segment lengths) and numerical values is key to solving the problem.

3. Extracting mathematical information from (diagrams) (Point 6)

- **Analysis:** Students must extract the implicit mathematical information from the provided diagram. For example, identifying that $150\mathrm{m}$ is the vertical distance from the highest point R to the river bed, and that $10\mathrm{m}$ is the distance from the boarding platform to the river bed. Converting the visual image into calculable mathematical lengths is an important step in modeling.

4. Perform a simple calculation (Point 1)

- **Analysis:** The final step to the solution involves basic arithmetic (subtraction or addition). Although the steps are simple (e.g., \(150 - 70 = 80\)), this is a necessary action to derive the final quantitative answer.

5. Make sense of, relate and use a variety of representations... (Point 10)

- **Analysis:** The problem combines a real-world picture, a geometric diagram, and a text description. Students need to connect these different representations: correlating the "center M" in the text with the point in the diagram, and the "highest point" with the value $150\mathrm{m}$, thus building a complete spatial model.

6. Use an understanding of the context to guide... the solving process (Point 13)

- **Analysis:** The physical context of a Ferris wheel helps students verify their results. For example, the center M should clearly be located between the highest and lowest points. The context guides students to realize they need to use the diameter to find the radius and subsequently determine the center's position.

1. Understand and utilise constructs based on definitions... (点 5)

- **分析：** 这是解题的核心。学生必须理解几何定义（直径与半径的关系）以及它们在物理空间中的规则。要找到中心 M 的高度，学生需要利用"半径等于直径的一半"这一基本规则（$140 \div 2 = 70$），并将其应用于给定的空间结构中。 (p. 1)

2. Manipulate numbers, graphical data... and geometric representations (点 7)

- **分析：** 题目提供了一个几何图形（圆、切线、垂直高度）。学生需要操作这些图形数据：从最高点高度（150m）减去半径（70m），或者将最低点高度（$150 - 140 = 10\text{m}$）加上半径（70m）。这种在几何表示（线段长度）和数值之间的操作是解题的关键。 (p. 1)

3. Extracting mathematical information from (diagrams) (点 6)

- **分析：** 学生必须从给出的示意图中提取隐含的数学信息。例如，识别出 150m 是从最高点 R 到河床的垂直距离，而 10m 是平台到河床的距离。将视觉图像转化为可计算的数学长度是建模的重要一步。 (p. 1)

4. Perform a simple calculation (点 1)

- **分析：** 最终的解决步骤涉及基础算术（减法或加法）。虽然步骤简单（如 $150 - 70 = 80$），但它是得出最终定量答案的必要行动。 (p. 1)

5. Make sense of, relate and use a variety of representations... (点 10)

- **分析：** 题目结合了实物图、几何示意图和文字描述。学生需要将这些不同的表现形式（representations）联系起来：将文字中的"中心 M"与图中的点对应，将"最高点"与数值 150m 对应，从而构建出完整的空间模型。 (p. 1)

6. Use an understanding of the context to guide... the solving process (点 13)

- **分析：** 摩天轮的物理背景帮助学生验证结果。例如，中心 M 显然应该处于最高点和最低点之间。背景知识引导学生意识到需要通过直径来寻找半径，进而确定圆心的位置。 (p. 1)